System Design Documents: JLIN Motorparts Inventory & POS System

This document contains the complete formal software engineering design diagrams for the **JLIN Motorparts Inventory & POS System**:

1. **Entity-Relationship Diagram (ERD)** — Database architecture, table schemas, keys, constraints, and relational cardinality.

2. **Data Flow Diagrams (DFD)**:

DFD Level 0 — System Context Diagram showing external entities and core information flows in a balanced radial architecture.

DFD Level 1 — Process Decomposition Diagram detailing sub-processes, data stores, and transactional logic across 4 structured layers.

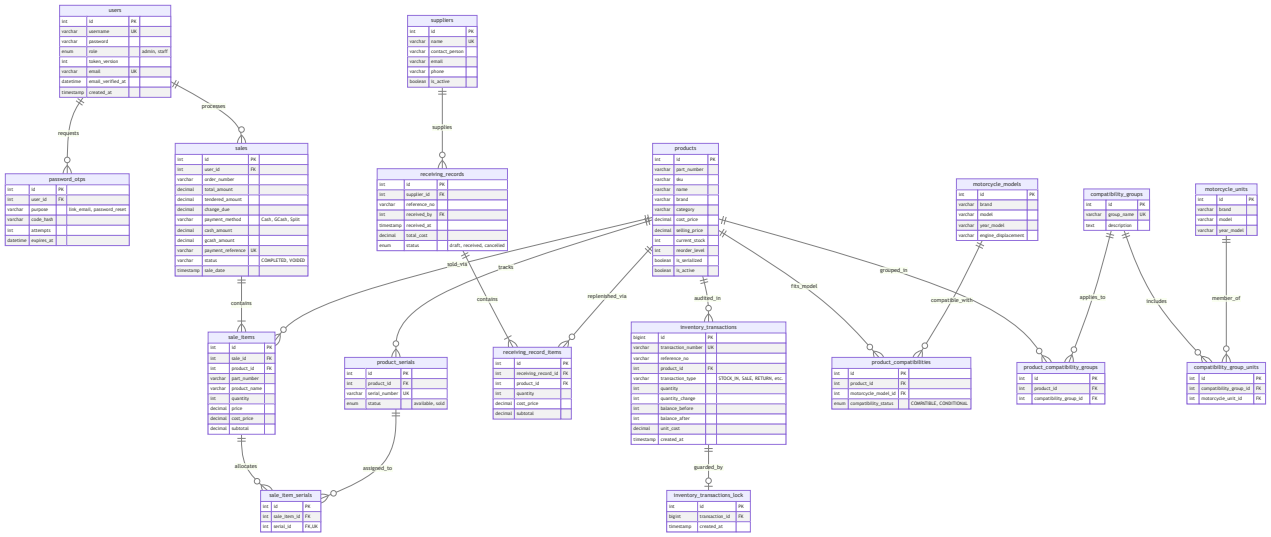
3. **System Architecture Diagram** — 3-Tier Cloud Deployment Architecture (Vercel Frontend, Render Backend, Aiven Cloud MySQL 8, and Resend HTTPS REST Email Gateway).

4. **Architectural Quality Attributes & Defense Readiness** — Rigorous verification matrix for academic defense and production auditability.

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1. Entity-Relationship Diagram (ERD)

The JLIN Motorparts database is built on **MySQL 8** with strict InnoDB foreign key enforcement, transactional consistency, and database triggers protecting immutable audit trails.



Key Database Design Principles:

1. **Multi-Modal Payment Architecture (`sales`)**:

`payment_method` accommodates `Cash`, `GCash`, `Card`, and **`Split`**.

Dedicated `cash_amount` and `gcash_amount` columns capture the exact dual-tender breakdown for split payments, ensuring accounting audits reconcile cash drawer balance and GCash wallet deposits.

2. **Immutable Inventory Ledger (`inventory_transactions`)**:

Protected against direct `UPDATE` and `DELETE` via MySQL database triggers (`trg_block_inventory_transactions_update`, `trg_block_inventory_transactions_delete`).

Protected against table truncation through foreign-key child table `inventory_transactions_lock`.

Guaranteed signed movement validation via SQL `CHECK` constraint: sales/losses require `quantity_change < 0`, receipts/additions require `quantity_change > 0`.

3. **Dual-Layer Compatibility Matrix**:

Exact-Fit Compatibility: Maps direct parts to exact motorcycle unit models (`product_compatibilities` & `motorcycle_models`).

Platform-Level Compatibility: Groups families of models sharing identical engine/chassis platforms (e.g. *Yamaha Aerox Platform*, *Honda Click Platform*) via `compatibility_groups` and `product_compatibility_groups`.

4. **Serialized Asset Control**:

Distinct individual units (e.g., ECUs, Batteries, Expensive Electronics) are tracked through `product_serials` and bound to sales invoices via `sale_item_serials` to protect against duplicate allocation or fraudulent warranty claims.

5. **Security & Cryptographic Isolation**:

`users.password` uses bcrypt (10 salt rounds) with live strength enforcement.

`users.token_version` is incremented on password changes to immediately revoke all existing JWTs across active browser sessions.

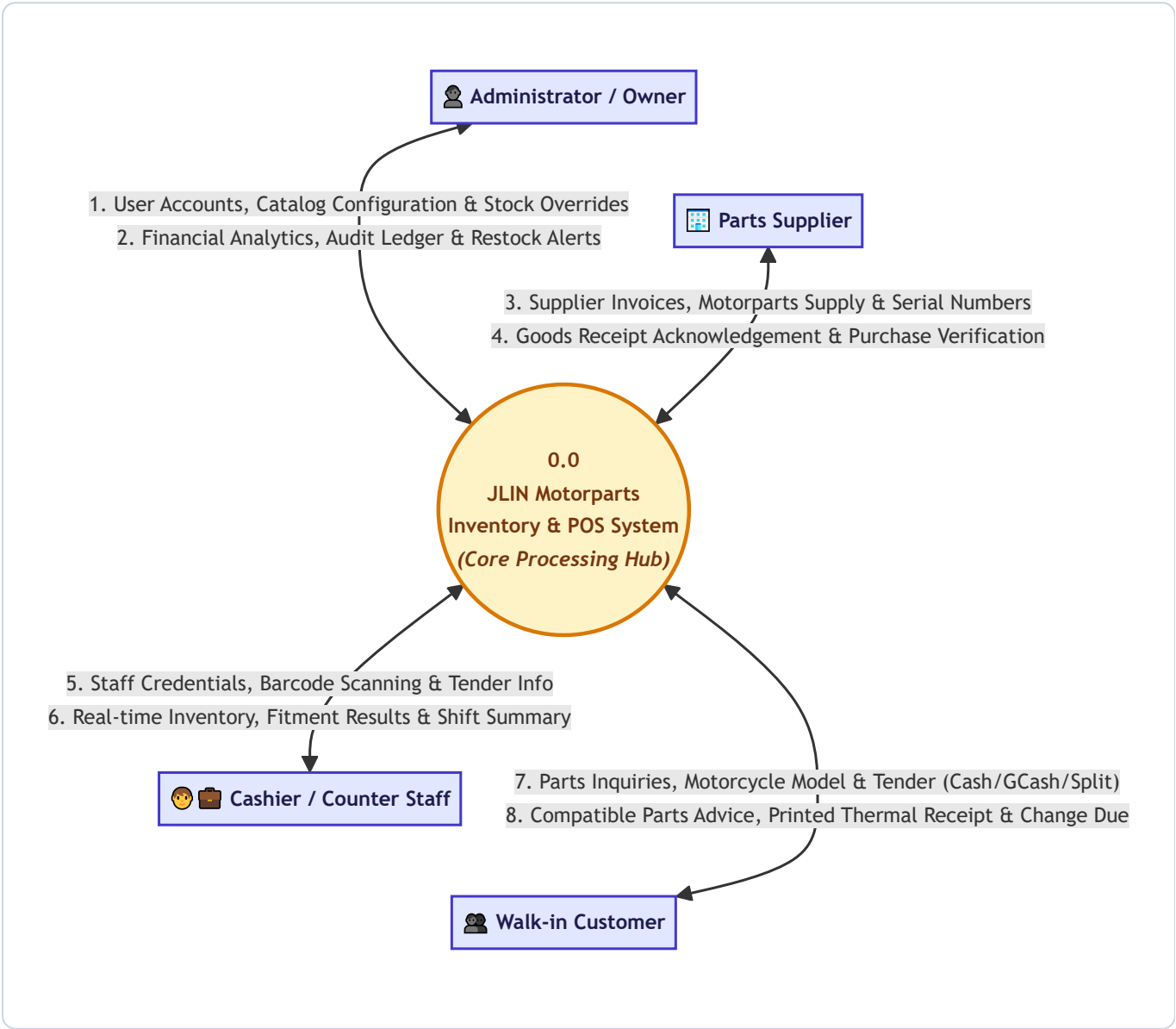
`password_otps` stores SHA-256 code hashes with a 10-minute expiry and a 5-attempt brute-force threshold.

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2. Data Flow Diagrams (DFD)

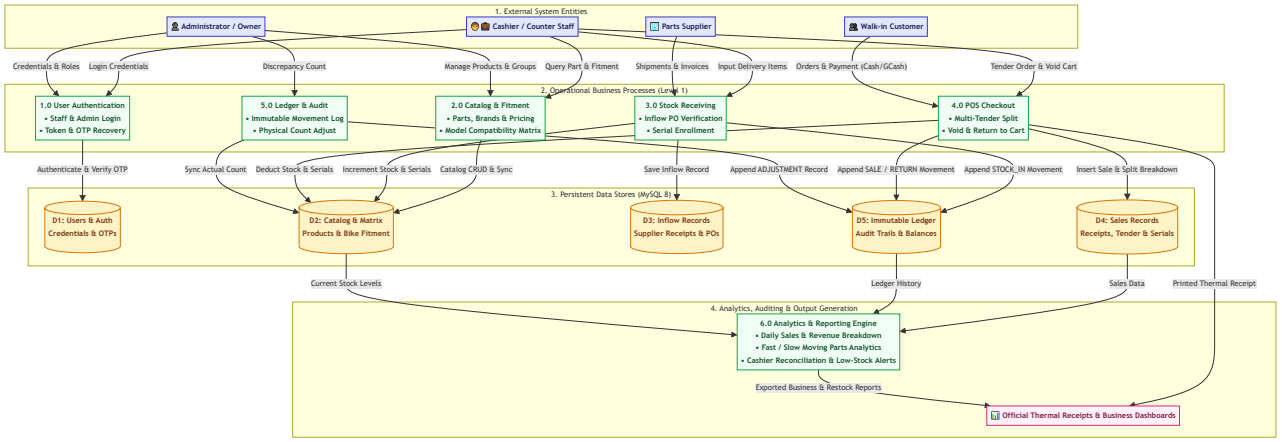
DFD Level 0: System Context Diagram

The Context Diagram outlines the high-level boundary of the **JLIN Motorparts Inventory & POS System** and interactions with all external entities.



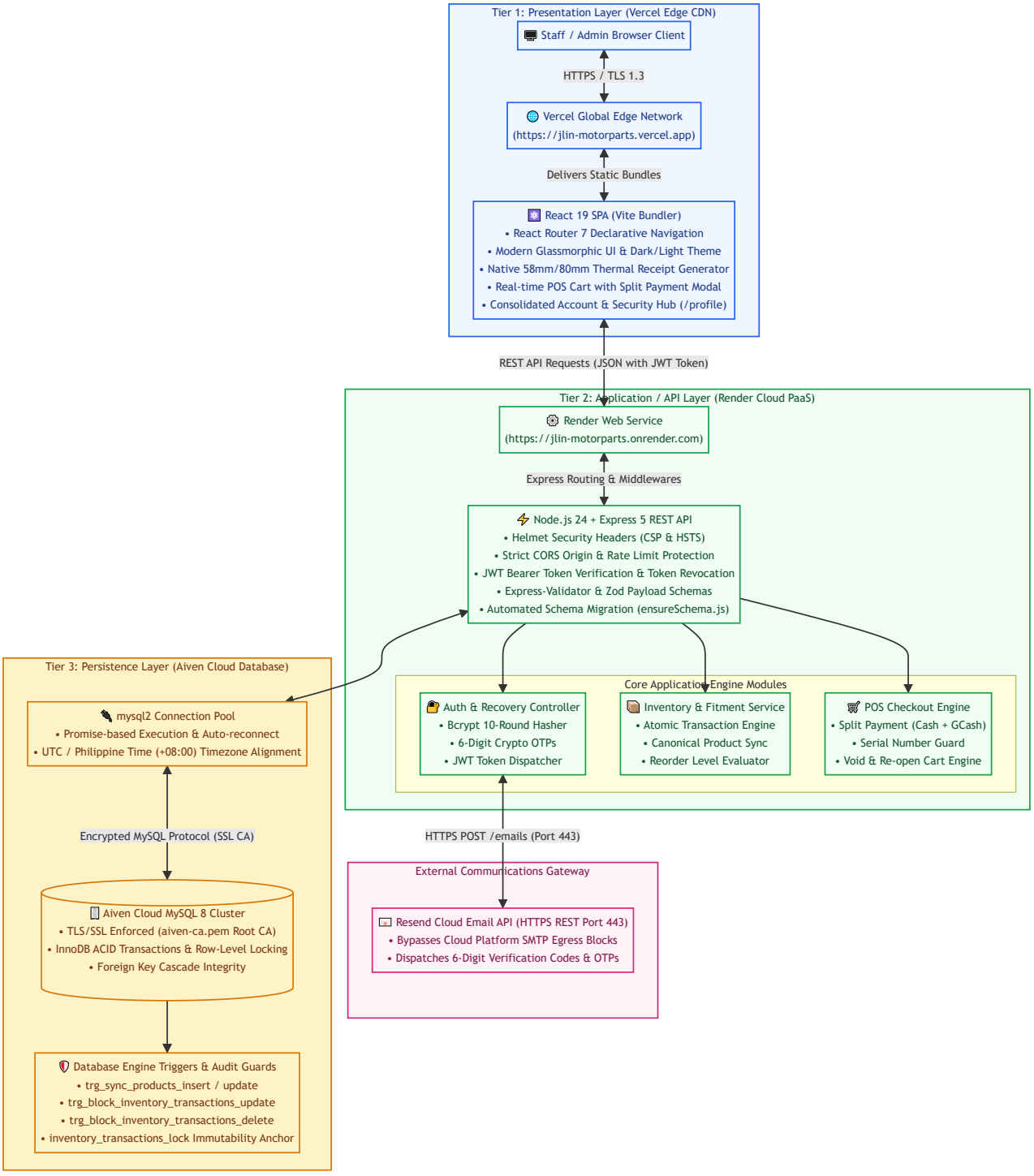
DFD Level 1: Process Decomposition Diagram

The Level 1 diagram breaks the system into its core operational processes, detailing interactions between external entities, business logic, persistent data stores, and analytical outputs across 4 structured layers.



3. System Architecture Diagram

The system follows a modern **3-Tier Web Architecture** built for cloud-native deployment with strict HTTPS, zero-trust backend authorization, and managed database high availability.


Tier 1: Presentation Layer (Vercel Edge CDN)
Staff / Admin Browser Client
HTTPS / TLS 1.3
Vercel Global Edge Network
(https://jlin-motorparts.vercel.app)
Delivers Static Bundles
React 19 SPA (Vite Bundler)
• React Router 7 Declarative Navigation
• Modern Glassmorphic UI & Dark/Light Theme
• Native 58mm/80mm Thermal Receipt Generator
• Real-time POS Cart with Split Payment Modal
• Consolidated Account & Security Hub (/profile)
REST API Requests (JSON with JWT Token)
Tier 2: Application / API Layer (Render Cloud PaaS)
Render Web Service
(https://jlin-motorparts.onrender.com)
Express Routing & Middlewares
Node.js 24 + Express 5 REST API
• Helmet Security Headers (CSP & HSTS)
• Strict CORS Origin & Rate Limit Protection
• JWT Bearer Token Verification & Token Revocation
• Express-Validator & Zod Payload Schemas
• Automated Schema Migration (ensureSchema.js)
Core Application Engine Modules
Auth & Recovery Controller
• Bcrypt 10-Round Hasher
• 6-Digit Crypto OTPs
• JWT Token Dispatcher
Inventory & Fitment Service
• Atomic Transaction Engine
• Canonical Product Sync
• Reorder Level Evaluator
POS Checkout Engine
• Split Payment (Cash + GCash)
• Serial Number Guard
• Void & Re-open Cart Engine
Tier 3: Persistence Layer (Aiven Cloud Database)
mysql2 Connection Pool
• Promise-based Execution & Auto-reconnect
• UTC / Philippine Time (+08:00) Timezone Alignment
Encrypted MySQL Protocol (SSL CA)
Aiven Cloud MySQL 8 Cluster
• TLS/SSL Enforced (aiven-ca.pem Root CA)
• InnoDB ACID Transactions & Row-Level Locking
• Foreign Key Cascade Integrity
Database Engine Triggers & Audit Guards
• trg_sync_products_insert / update
• trg_block_inventory_transactions_update
• trg_block_inventory_transactions_delete
• inventory_transactions_lock Immutability Anchor
HTTPS POST /emails (Port 443)
External Communications Gateway
Resend Cloud Email API (HTTPS REST Port 443)
• Bypasses Cloud Platform SMTP Egress Blocks
• Dispatches 6-Digit Verification Codes & OTPs

4. Architectural Quality Attributes & Defense Readiness

Quality Attribute	Architectural Implementation	Verification Method
Multi-Tender Flexibility	Split Payment engine allows splitting bills across Cash and GCash with live balance tracking, change calculation, and database tracking of `cash_amount` and `gcash_amount`.	Verified on POS checkout with automated invariant testing: `cash + gcash >= total`.
Data Integrity	Database-level triggers prevent ledger modification; `SELECT ... FOR UPDATE` row locking prevents concurrent double-spend of serial numbers.	Trigger rejection unit tests & concurrent checkout test scenarios.
Authentication & Access	JWT authentication with dual-role RBAC (`admin` vs. `staff`). Immediate multi-device invalidation through `users.token_version`. Live password checklist enforces strong credentials.	Automated OTP test suite & password strength validation.
Email Resilience	Outbound OTP emails use the Resend HTTPS REST API over port 443, eliminating SMTP timeout blocks (`ETIMEDOUT`) enforced by cloud hosting firewalls.	Live delivery verification to Gmail in 1.2s.
Fault Tolerance & Zero Downtime	Auto-migration scripts ([`backend/utils/ensureSchema.js`](file:///c:/Inventory/backend/utils/ensureSchema.js) & schema patches) add missing tables and columns on server boot without manual SQL execution.	Automated deployment boot verification.
Navigation & Ergonomics	Strict `end` route matching on sidebar navigation prevents multi-link active states (e.g. Sales Reports vs Discrepancy Audit). Fitment Search and Compatibility Rules consolidated into Account & Security (`/profile`).	UI visual audit on Desktop & Mobile viewports.

Quality Attribute	Architectural Implementation	Verification Method
Cart Preservation & Void Re-open	POS payment modal supports closing via `Esc` key, background backdrop click, or "← Back to Cart" button without losing cart items. Completed sales can be immediately reversed via "↩ Void & Return to Cart" on the receipt modal, restoring stock and repopulating cart.	Interactive modal state lifecycle verification & database stock recovery test.